

Semantic Theory 2026: Practice Exam Key

Question 1: Predicate Logic (10)

a. Translate the following sentences into first-order predicate logic:

i. *not all students turn in every assignment*

either of the following:

$$\neg \forall x [student(x) \rightarrow \forall y [assignment(y) \rightarrow turn-in(x, y)]]$$

$$\forall y [assignment(y) \rightarrow \neg \forall x [student(x) \rightarrow turn-in(x, y)]]$$

ii. *Mary doesn't see everything that she hears*

$$\neg \forall x [hear(m', x) \rightarrow see(m', x)]$$

Question 2: Type Theory (20)

Provide the derivations (type inferencing) of each of the following sentences. Brackets indicate the combinatorics, and subscripts indicate the types of (some of) the expressions—the rest must be deduced. You can treat “*the ship*” as the single term s' in (b).

a. $[some\ dog] [ate_{\langle e, \langle e, t \rangle \rangle} [a\ biscuit]]$

$$\frac{\frac{some :: \langle et, \langle et, t \rangle \rangle \quad dog :: et}{some(dog) :: \langle et, t \rangle} \quad \frac{\frac{a :: \langle et, \langle \langle e, et \rangle, et \rangle \rangle \quad biscuit :: et}{a(biscuit) :: \langle \langle e, et \rangle, et \rangle} \quad eat :: \langle e, et \rangle}{a(biscuit)(eat) :: et}}{some(dog)(a(biscuit)(eat)) :: t}$$

b. $[every\ gardener] [likes_{\langle et, t \rangle} potatoes_e]$

$$\frac{\frac{every :: \langle et, \langle et, t \rangle \rangle \quad gardener :: et}{every(gardener) :: \langle et, t \rangle} \quad \frac{like :: \langle e, et \rangle \quad p' :: e}{like(p') :: et}}{every(gardener)(like(p')) :: t}$$

Question 3: λ -Calculus (10)

a. Given the types that you determined (or were given) for the terms in Question 2, derive the corresponding λ -expressions for the following terms:

- i. *likes* $\lambda x_e \lambda y_e. like^*(x)(y)$
- ii. *some* $\lambda P_{et} \lambda R_{et}. \exists x [P(x) \wedge R(x)]$
- iii. *a* $\lambda P_{et} \lambda R_{\langle e, et \rangle} \lambda y_e. \exists x [P(x) \wedge R(x)(y)]$

Question 4: Generalized Quantifiers (20)

Consider the following sentence: *Only George can solve the problem*

a. Give the generalized quantifier definition of the noun phrase “*only George*”.

$$\llbracket \text{only George} \rrbracket^M = \{P \subseteq U_M \mid |P| = 1 \wedge \llbracket \text{George} \rrbracket^M \in P\}$$

$$\text{OR: } \llbracket \text{only George} \rrbracket^M = \{P \subseteq U_M \mid P = \{\llbracket \text{George} \rrbracket\}\}$$

b. What are the monotonicity properties (left and right) of *only*? Show how you derived these properties.

Left upward:

$$\text{only good students came to the party} \models \text{only students came to the party}$$

Right downward:

$$\text{only athletes ran} \models \text{only athletes ran quickly}$$

Question 5: Event Semantics (20)

a. Translate the following sentences into Davidsonian (event semantics) representations, **including** temporal information.

i. *a woman witnessed a robbery on Saturday*

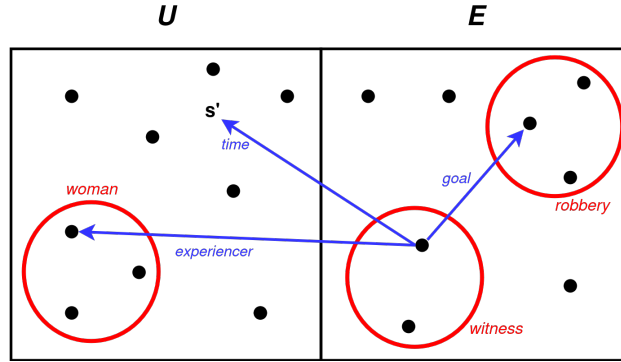
$$\exists x \exists e_1 \exists e_2 [\text{witness}(e_1) \wedge \text{woman}(x) \wedge \text{exp}(e_1, x) \wedge \text{robbery}(e_2) \wedge \text{goal}(e_1, e_2) \wedge e_1 < e_u \wedge \text{time}(e_1, s')]$$

ii. *she_e she saw a burglar taking a watch*

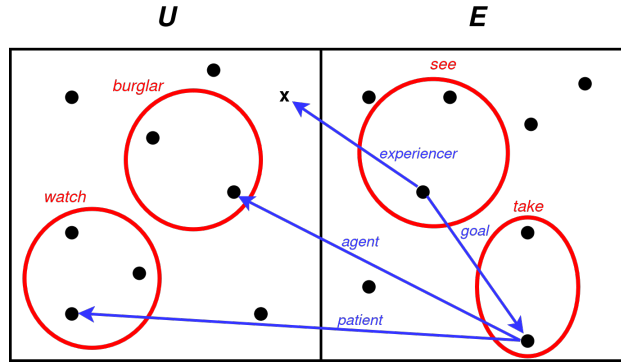
$$\exists y \exists z \exists e_1 \exists e_2 [\text{see}(e_1) \wedge \text{exp}(e_1, x) \wedge \text{take}(e_2) \wedge \text{goal}(e_1, e_2) \wedge e_1 < e_u \wedge \text{burglar}(y) \wedge \text{ag}(e_2, y) \wedge \text{watch}(z) \wedge \text{pat}(e_2, z)]$$

b. For one of the sentences in (a) (you pick), draw a Davidsonian model structure in which the sentence holds. You may ignore the temporal aspects of the sentences here.

i.



ii.



Question 6: Lexical Semantics (20)

Consider the following sentence: $S = \text{"two students ate a pizza"}$

a. How many readings does the sentence S have? List all possible readings in natural language.

Let:

D_s = the distributive reading of "two students"

C_s = the collective reading of "two students"

We have the following readings:

- i. $\exists > C_S$: two students ate the same pizza together
 - ii. $\exists > D_S$: two students ate the same pizza, but not together
 - iii. $C_S > \exists$: same as (i)
 - iv. $D_S > \exists$: two students separately ate a pizza (not necessarily the same pizza)
- b. Translate each reading of S to the extended first-order logic for plural terms, where variables $X, Y, Z, \dots$ range over proper sums, $X \oplus Y$ denotes the group consisting of X and Y , $\triangleleft$ denotes the part-of-relation, and $N(X) = |\{y \mid At(y) \wedge y \triangleleft X\}|$ takes a proper sum X and returns the number of atoms in X .

- i. $(\exists > C_S): \exists x \exists Y [pizza_{SG}(x) \wedge student_{PL}(Y) \wedge N(Y) = 2 \wedge eat(Y, x)]$
- ii. $(\exists > D_S): \exists x \exists Y [pizza_{SG}(x) \wedge student_{PL}(Y) \wedge N(Y) = 2 \wedge \forall z [At(z) \wedge z \triangleleft Y \rightarrow eat(z, x)]]$
- iii. $(C_S > \exists)$: same as (i)
- iv. $(D_S > \exists): \exists Y [student_{PL}(Y) \wedge N(Y) = 2 \wedge \forall z [At(z) \wedge z \triangleleft Y \rightarrow \exists x [pizza_{SG}(x) \wedge eat(z, x)]]]$

Question 7: Dynamic Semantics (10)

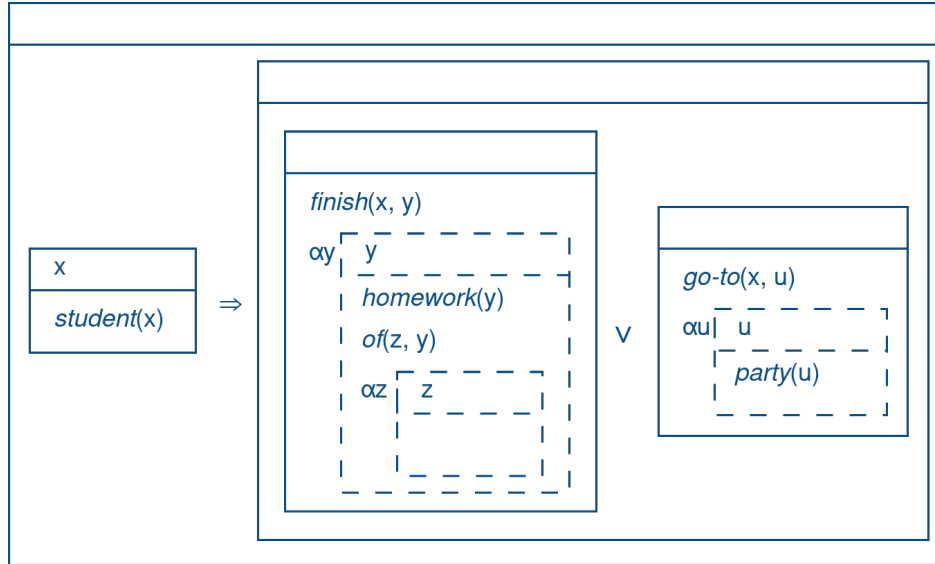
Translate the following natural language utterances into Dynamic Predicate Logic..

- i. *If Mary_i eats a sandwich, she_i is happy. She_i ate a sandwich_k. It_k was delicious.*
 $((\exists x \cdot sandwich(x) \cdot eat(m', x)) \rightarrow happy(m')) \cdot (\exists y \cdot sandwich(y) \cdot eat(m', y)) \cdot delicious(y)$
 $= \sim(\exists x \cdot sandwich(x) \cdot eat(m', x) \cdot \sim happy(m')) \cdot \exists y \cdot sandwich(y) \cdot eat(m', y) \cdot delicious(y)$
- ii. *Everyone who eats a sandwich drinks a beer.*
 $(\exists x \cdot person(x) \cdot \exists y \cdot sandwich(y) \cdot eat(x, y)) \rightarrow (\exists z \cdot beer(z) \cdot drink(x, z))$
 $= \sim(\exists x \cdot person(x) \cdot \exists y \cdot sandwich(y) \cdot eat(x, y) \cdot \sim(\exists z \cdot beer(z) \cdot drink(x, z)))$

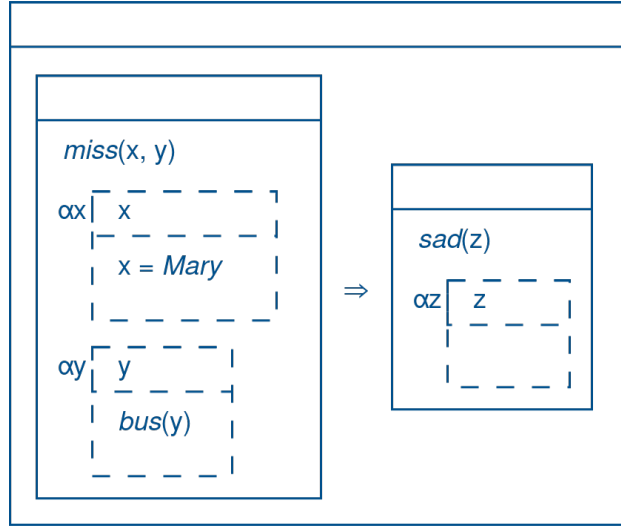
Question 8: DRT and Presuppositions (30)

- a. Give proto-DRSs for the following sentences. Treat “goes to” in (i) as the two-place predicate *go-to*(*x*, *y*):

- i. *Every student finishes their homework or goes to the party*

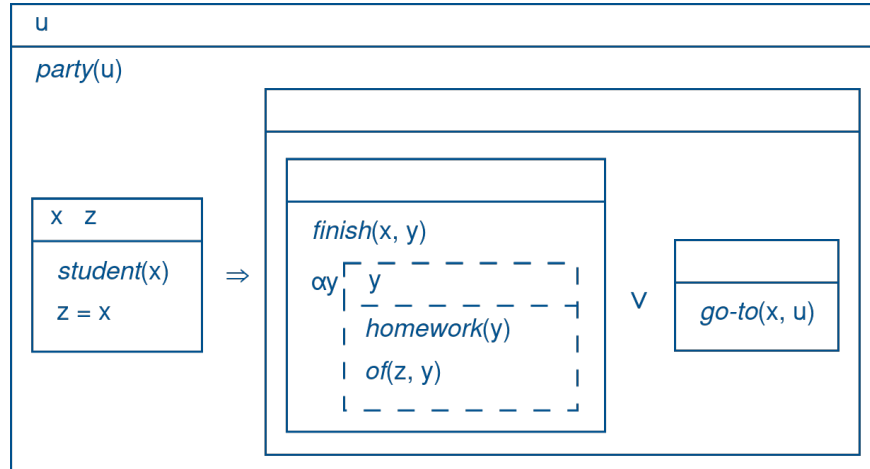


- ii. *If Mary misses the bus, she is sad*

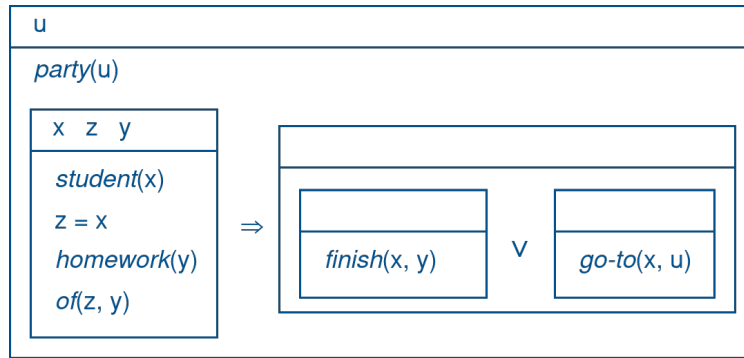


- b. Resolve the proto-DRSs in (a). Explicitly describe the resolution constraints you apply.

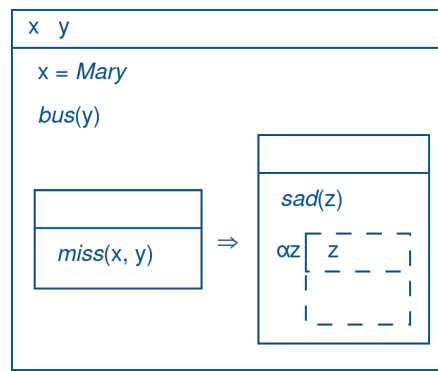
- i. We cannot resolve αy yet, because it is not α -free: we start with αz and αu . There is no possible anaphor that αu can bind to, so we accommodate. The highest-possible DRS is preferred for accommodation, so we accommodate at the top-level DRS. We do have a possible anaphor for αz (*student*), so we bind at the left-hand side of $\Rightarrow$ (the closest possible anaphor).



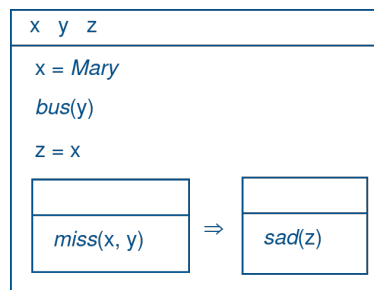
Now αy is α -free, so we can resolve it. There is no possible anaphor, so we must accommodate at the highest-possible DRS. Accommodation at the top-level DRS would violate the free variable constraint (it would leave z free), so we accommodate at the next-highest DRS: the left-hand side of $\Rightarrow$.



- ii. We can't resolve αz yet, because the target anaphor (*Mary*) hasn't been resolved: we start with αx and αy . Neither has a possible anaphor, so we accommodate at the top-level DRS.



Now we can resolve αz by binding to the closest possible anaphor (x), which is in the top-level DRS.



- c. Give the truth-conditions for one of the **resolved** DRSs in (b) (you pick). Use verifying embeddings to arrive at the model-theoretic interpretation.

A DRS $K = (U_K, C_K)$ is true in a model $M = (U_M, V_M)$ iff there exists $f: U_D \rightarrow U_M$ such that:

- $U_K \subseteq \text{dom}(f)$
- f verifies K in M ($f \models_M K$)

i. Let:

- $K = (\{u\}, \{\text{party}(u), K_1 \Rightarrow K_2\})$
- $K_1 = (\{x, y, z\}, \{\text{student}(x), z = x, \text{homework}(y), \text{of}(z, y)\})$
- $K_2 = (\emptyset, \{K_3 \vee K_4\})$
- $K_3 = (\emptyset, \{\text{finish}(x, y)\})$
- $K_4 = (\emptyset, \{\text{go-to}(x, u)\})$

Then $f \models_M K$ iff:

- $f(u) \in V_M(\text{party})$
- and $f \models_M (K_1 \Rightarrow K_2)$:
for all g_1 , if:
 - $g_1 \supseteq_{\{x, y, z\}} f$: $g_1(u) = f(u)$ (and $\{u, x, y, z\} \subseteq \text{dom}(g_1)$)
 - and $g_1 \models_M K_1$: $g_1(x) \in V_M(\text{student})$, $g_1(z) = g_1(x)$, $g_1(y) \in V_M(\text{homework})$, and $(g_1(z), g_1(y)) \in V_M(\text{of})$
- ... then there exists g_2 such that:
 - $g_2 \supseteq_{\emptyset} g_1$: $g_2(u) = g_1(u)$, $g_2(x) = g_1(x)$, $g_2(y) = g_1(y)$, and $g_2(z) = g_1(z)$
 - and $g_2 \models_M K_2$:
 - * there exists g_3 such that:
 - $g_3 \supseteq_{\emptyset} g_2$: $g_3(u) = g_2(u)$, $g_3(x) = g_2(x)$, $g_3(y) = g_2(y)$, and $g_3(z) = g_2(z)$
 - and $g_3 \models_M K_3$: $(g_3(x), g_3(y)) \in V_M(\text{finish})$
 - * OR there exists g_4 such that:
 - $g_4 \supseteq_{\emptyset} g_2$: $g_4(u) = g_2(u)$, $g_4(x) = g_2(x)$, $g_4(y) = g_2(y)$, and $g_4(z) = g_2(z)$
 - and $g_4 \models_M K_4$: $(g_4(x), g_4(u)) \in V_M(\text{go-to})$

ii. Let:

- $K = (\{x, y, z\}, \{x = \text{Mary}, \text{bus}(y), z = x, K_1 \Rightarrow K_2\})$
- $K_1 = (\emptyset, \{\text{miss}(x, y)\})$
- $K_2 = (\emptyset, \{\text{sad}(z)\})$

Then $f \models_M K$ iff:

- $f(x) = V_M(\text{mary})$, $f(y) \in V_M(\text{bus})$, $f(z) = f(x)$
- and $f \models_M (K_1 \Rightarrow K_2)$:
for all g_1 , if:
 - $g_1 \supseteq_{\emptyset} f$: $g_1(x) = f(x)$, $g_1(y) = f(y)$, and $g_1(z) = f(z)$
 - and $g_1 \models_M K_1$: $(g_1(x), g_1(y)) \in V_M(\text{miss})$
- ... then there exists g_2 such that:
 - $g_2 \supseteq_{\emptyset} g_1$: $g_2(x) = g_1(x)$, $g_2(y) = g_1(y)$, and $g_2(z) = g_1(z)$
 - $g_2 \models_M K_2$: $g_2(z) \in V_M(\text{sad})$